

Ausgestaltung eines Lehrkonzepts für Grundlagenveranstaltungen zur Softwareentwicklung in Zeiten von Generative Artificial Intelligence (GenAI)

*Development of a Teaching Concept for Introductory Courses in
Software Development in the Age of Generative Artificial Intelligence*

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NOTE: This version of the Exposé is AI-assisted and CANNOT be officially submitted. It is for clarification purposes with the professor Boettcher only

Zusammenfassung

Generative Artificial Intelligence (GenAI) rapidly transforms both industry and academia. Teachers see didactical potential as well as challenges. This thesis explores how a curriculum for introductory software development courses need to be adjusted without risking constrictive alignment or academic integrity and set new learning goals.

The work begins with an introduction that explains the necessity of further exploration of this topic and defines the research questions. After the introduction, the theoretical framework summarizes the current literature status of AI in higher education, discusses important didactical concepts, explores challenges and potential of AI usage in higher education and analyses the current usage status of GenAI at the university of applied sciences Munich.

A curriculum analysis follows, identifying the current learning goals, examinations formats, constructive alignment, evaluation and challenges of integrating AI.

An empirical study done by questionnaires among a sample of students reflecting on their usage, winnings and concern of using GenAI during attending SE1. Interviews amongst students of the "Fachschaft" who already passed SE1 will be questioned about how GenAI affected their programming skills after passing the class.

In the conclusion, we will summarize the findings of the study to form recommended adjustments to the current curriculum in a way that benefits students and doesn't harm academic integrity.

Outline

1. Introduction

- Motivation
- Research Questions

2. Theoretical Framework

- Literature Review: Current Status of Research on Integrating AI in Higher Education
- Teaching Concepts (e.g., Constructive Alignment)
- Current Usage Status of GenAI at university of applied sciences munich (HM) (Vibe Coding, Over-reliance)

3. Curriculum Analysis

- Learning Goals
- Examination Formats and Constructive Alignment
- Evaluation and Challenges

4. Empirical Study

- Questionnaire
- Interviews

5. Discussion

- Contribution of Findings to Curriculum Development

6. Conclusion

1 Introduction

Generative Artificial Intelligence (GenAI) has become an unavoidable force in both academia and industry. In software development education, particularly in introductory courses like Softwareentwicklung 1 (SE1), its integration poses both opportunities as well as concerns. This thesis explores how GenAI can be meaningfully embedded into the curriculum while maintaining pedagogical integrity.

2 Motivation

The rise of GenAI tools such as ChatGPT and GitHub Copilot has transformed the expectations placed on software developers. In professional environments, programmers are increasingly expected not only to write code but to leverage AI tools to enhance productivity. This shift is mirrored in educational contexts, where students often adopt GenAI tools more readily than their instructors.

This creates a dual challenge: on one hand, educators must address concerns about academic dishonesty, plagiarism, and over-reliance on AI; on the other, they must recognize the potential of GenAI to reduce repetitive tasks, offer personalized tutoring, and expand access to learning resources. Most instructors agree that banning AI outright is neither practical nor pedagogically sound. Instead, this thesis aims to help educators prepare students to use GenAI responsibly, understand its limitations, and define clear expectations for AI-assisted and independent work.

3 Research Questions

The initial Research Questions will be used for the theoretical framework. These were:

- What are the motivations, benefits and risks of students using AI?
- What are the motivations, benefits and risks of instructors integrating AI?
- How is the current curriculum of Hochschule München in Softwareentwicklung 1 and what aspects need to be adjusted for AI integration?
- What are the new learning goals that are expected from students, compared to what was expected prior?
- How can we limit risks and assure academic integrity/prevent plagiarism?
- Exams that forbid the usage of AI cause issues with constructive alignment of what is taught and what is examined

These research questions help us understand what the current consensus of students and instructors regarding AI is, how the curriculum could be adjusted and what we expect from students that have the ability to use AI

The newly presented research questions from prof. boettcher's thoughts are:

- Does the exclusion of AI chatbots in exams violate constructive alignment? -> questionnaire, examination formats
- How can AI be used in an educational context for an introductory software-development course?

3.1 *Constructive Alignment and the Assessment Dilemma*

- How does the embedding of AI assistance impact the curriculum and what shifts need to be made?
- How should learning goals be re-aligned with the new curriculum?
- How should instructors examine students and what competences should instructors expect from students (with and without AI assistance)?

3.1 Constructive Alignment and the Assessment Dilemma

A central tension arises from the constructive alignment principle: if GenAI is taught but excluded from exams (as currently planned for SE1 on EXaHM), the alignment between learning objectives, teaching methods, and assessment is disrupted. While this may be pragmatically necessary, it invites critical reflection. Can we justify this misalignment pedagogically? What alternative assessment strategies could preserve integrity while acknowledging GenAI's role?

3.2 Dual Role of GenAI in Education

Two complementary directions for GenAI integration are identified:

1. **GenAI as a Learning Support Tool:** Acting as a personal tutor, GenAI can guide students through coding exercises, explain concepts, and offer feedback. Studies such as Bassner et al. [?] demonstrate how LLMs can be adapted with reasonable effort to support learning.
2. **GenAI as a Software Development Assistant:** Tools like GitHub Copilot and more advanced models assist in code generation, debugging, and documentation. Their use reflects real-world practices and should be mirrored in educational settings.

3.3 Learning Objectives

To align teaching with the presence of GenAI, new learning goals must be defined. These include:

- Understanding the capabilities and limitations of GenAI tools.
- Developing critical thinking and verification strategies when using AI-generated code.
- Ethical awareness of plagiarism, authorship, and academic integrity.
- Ability to distinguish between AI-assisted and independently acquired knowledge.

3.4 Curriculum Adaptation

The current SE1 curriculum must be reviewed to identify areas where GenAI can be integrated. This includes:

- Revision of lecture content to include GenAI use cases.
- Design exercises that encourage AI-assisted exploration while requiring human reasoning.
- Creating guidelines for responsible AI usage in assignments.

3.5 Teaching and Learning Activities

Inspired by the above papers, the following formats are proposed:

- **Inverted Classroom:** Students prepare with GenAI-assisted materials and engage in critical discussions during class.
- **Practical Labs:** Structured tasks in which students use GenAI tools under supervision, followed by reflection on their process and outcomes.

3.6 Examination Strategy

Although SE1 exams currently exclude GenAI, this thesis will explore alternative formats such as:

- Split assessments: AI-assisted coursework vs. AI-free exams.
- Oral exams focusing on the reasoning behind AI-generated solutions.
- Code reviews in which students justify and critique AI-generated code.

4 Theoretical framework

explanation on genAI, overview on genAI tools, teaching concepts e.g. constructive alignment, current curriculum + usage of AI in HM, examination at HM

- genAI: explanation, overview of tools
- didactic: constructive alignment, current curriculum and analysis, AI usage, examination of students, learning goals

5 Empirical Study

- Methodology: interviews or surveys with students and educators
- Implementation and analysis
- Insights into usage, acceptance, and criticism of GenAI

6 Curriculum Analysis and Adaptation

- Analysis of the current SE1 curriculum
- Suggestions for integrating GenAI into lectures and exercises
- Development of guidelines for responsible use

7 Teaching and Learning Formats

- Inverted classroom with GenAI-based preparation
- Practical exercises with reflection
- Promotion of critical thinking and personal responsibility

8 Examination Formats and Evaluation

- The dilemma of AI-free exams
- Suggestions: split assessment, oral exams, code reviews
- Ensuring academic integrity

9 Comparison with Other Universities

- Case studies from Germany and abroad
- Best practices and lessons learned
- Deriving recommendations for Hochschule München

10 Reflection and Outlook

- Personal and academic reflection on the role of GenAI
- Future perspectives for teaching and curriculum
- Recommendations for long-term development

11 Conclusion

- Summary of findings
- Answers to the research questions
- Implications for teaching, assessment, and higher education didactics

12 Expected Contribution

This thesis aims to provide a structured framework for integrating GenAI into SE1 at the Hochschule München. It will offer:

- A literature-based rationale for GenAI integration.
- A set of revised learning goals and teaching strategies
- Recommendations for the redesign of the curriculum and assessment.
- A critical discussion of constructive alignment in the age of AI.

1. Introduction

- Relevance of GenAI in higher education
- Research motivation and objectives
- Structure of the thesis

2. Educational Transformation through GenAI

- Shifting skill requirements for future developers
- Opportunities and risks of GenAI in learning environments
- Ethical and pedagogical implications

3. Research Framework

- Central research questions
- Methodological approach
- Connection to educational theory

4. GenAI as a Learning Companion

- Roles of GenAI: tutor, assistant, co-creator
- Literature review on GenAI in education
- Critical perspectives and controversies

5. Curriculum Innovation

- Analysis of current SE1 curriculum
- Integration strategies for GenAI
- Guidelines for responsible use

6. Teaching Formats and Learning Activities

- GenAI-supported inverted classroom
- Reflective practical exercises
- Promoting autonomy and critical thinking

7. Assessment and Integrity

- Challenges of AI-free exams
- Hybrid assessment models
- Safeguarding academic integrity

8. Comparative Case Studies

- Insights from other universities
- Best practices and implementation models
- Transferability to Hochschule München

9. Reflection and Future Outlook

- Personal and academic reflection
- Long-term implications
- Recommendations for sustainable integration

10. Conclusion

- Summary of findings
- Answering the research questions
- Implications for teaching and policy